

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 1st Semester B.Sc. in CSE Final Examination, 2021
CSE-1101: Fundamentals of Computers and Computing

Total Marks: 70

Time: 3 Hours

Answer any 5 (Five) of the following Questions

1.
 - a) Explain how read/write operation is carried out in an optical disk. 4
 - b) RAM is called the working memory of the computer. Expand and Explain. 4
 - c) What is the average access time for transferring 512 bytes of data with the following specifications: 4
 - Average seek time = 5 msec
 - Disk rotation = 6000 RPM
 - Data rate = 40 KB/sec
 - Controller overhead = 0.1 msec
 - d) Define Seek Time, Access Time and Rotational Latency in Disks 2
2.
 - a) What is Operating System? Explain the booting process of an operating system. 6
 - b) What is file system? What does a file system do? 4
 - c) How does a computer process data? 4
3.
 - a) Why Unicode is important? 3
 - b) Write down the full form of BCD, ASCII and EBCDIC codes 3
 - c) What are the differences between SRAM and DRAM? 4
 - d) Define data and information. Write down the names of five output devices. 4
4.
 - a) Write down the factors affecting the processing speed of a computer system. 4
 - b) What do you mean by RJ-45? Write short note on Star and Ring topologies. 4
 - c) Convert the following 6
 - i) $(3D59)_{16} = (?)_2$
 - ii) $(786)_{10} = (?)_8$
 - iii) $(111011)_2 = (?)_{10}$
5.
 - a) What is DNS? How does DNS work? 5
 - b) Why is the IP address called "logical" address, and the MAC address called "physical" address? 4
 - c) Find the error of the following IP addresses. If the IP is valid, check its Class and determine whether the address is Unicast or Multicast. 5
 - i) 237.15.2.1
 - ii) 256.1.8.9
 - iii) 114.34.2.8
 - iv) 255.254.0.1
 - v) 8.8.8.8
6.
 - a) Write a comparative study between impact printer and nonimpact printer. 4
 - b) Mention the main differences between HDD and SSD. 4
 - c) What are the basic functionalities of a Microprocessor? 4
 - d) Why ping and traceroute command is used in networking? 2
7.
 - a) Write a comparative study among Machine Language, Assembly Language and Higher Level Language. 6
 - b) What is variable? Describe the basic data types in C. 4
 - c) Write a program in C to find the largest among three whole decimal numbers. 4